Syed Shakeel Bacha

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Research Profile

Aspiring AI Researcher with a focused interest in **Natural Language Processing (NLP)**, **Deep Learning**, and **Generative AI**. Developed an end-to-end Medical AI ecosystem for my BS Thesis, achieving **98.12% accuracy** in tumor detection using **CNNs** and integrating **RAG-based Large Language Models (LLMs)** for clinical consultation. Leveraging 2 years of professional Web Development experience, I specialize in bridging the gap between advanced machine learning research and scalable deployments. Currently based in Riyadh and prepared for international relocation to South Korea.

Education

Abdul Wali Khan University Mardan (AWKUM) | *Mardan, Pakistan*
Bachelor of Science in Computer Science (Software) 2021 – 2025
CGPA: 3.15 / 4.0 | 78%
Relevant Coursework: Artificial Intelligence, Neural Networks, Computer Vision, Data Structures, Algorithms.

The Quaid-e-Azam School and College | *Mardan, Pakistan*
FSc (Pre-Engineering / Computer Science) 2019 – 2021
Marks: 824 / 1100 | 75%

Frontier Model School & College | *Toru, Mardan, Pakistan*
Matriculation (Science) 2017 – 2019
Marks: 839 / 1100 | 76%

Research Projects

Brain Tumor Detection System with AI Chatbot & Hospital Recommendation System *BS Thesis*
Awarded 2nd Position in the Final Year Project Exhibition 2025

- Engineered a **VGG16-based CNN** for multi-class classification, achieving **98.12%** accuracy on 7,033 MRI scans.
- Integrated **Grad-CAM** for Explainable AI (XAI), providing 92% concordance with radiologist annotations.
- Developed an **RAG-based Chatbot** using **Mistral 7B** and **LangChain** to query neuro-oncology papers.
- Designed a **Hospital Recommendation System** and deployed via a **Flask Web Application**.

Technical Skills

Programming Languages: **Python**, **C**, **SQL**, **Java**, **LaTeX**, JavaScript (ES6+)
AI/ML Frameworks: PyTorch, TensorFlow, HuggingFace, LangChain, OpenAI API, OpenCV
Web Technologies: React.js, Node.js, HTML5, CSS3, Tailwind CSS, Flask, FastAPI
Tools & Infrastructure: Git, Docker, Linux, **Networking Basics (Cisco)**, MongoDB, Oracle

Skills

Productivity Suite: **Microsoft Office (Word, Excel, PowerPoint)**, Google Drive, Skype, Zoom
Core Competencies: **Organizational and Planning Skills**, Technical Writing, Social Media Management
Research Engineering: RAG, Explainable AI (XAI), Full-stack Development, Troubleshooting
Management: Quality Improvement & Management, Project Documentation, System Administration

Professional Experience

Web Developer (Freelance/Contract) 2022 – 2024

- Designed and deployed responsive web applications using **React.js** and **Node.js** for 2+ years.
- Optimized web performance and SEO, focusing on high-performance UI/UX and RESTful API integration.

Tech Training Lab Jun 2024 – Aug 2024
IT Support / Database Trainee (Internship) Mardan, Pakistan

- Configured and managed **Oracle Database** environments; performed SQL-based data retrieval and schema operations.
- Leveraged **Linux** commands for system navigation, user management, and permission control.

Languages

- **English:** Professional Working Proficiency (Fluent in Research & Technical Writing)
- **Urdu:** Native
- **Pashto:** Native
- **Arabic:** Professional Working Proficiency (Riyadh, Saudi Arabia)

Certifications

- **Networking Basics** – Cisco Networking Academy
- **AI For Everyone** – Coursera (DeepLearning.AI)
- **Machine Learning Specialization** – Stanford University & DeepLearning.AI
- **Advanced Learning Algorithms** – DeepLearning.AI
- **Supervised Machine Learning** – DeepLearning.AI
- **Quality Improvement and Management** | **Document Design** – Coursera